

$$\begin{aligned}
\log p_\theta(x) &= \log \int p_\theta(x|z)p_\theta(z)dz \\
\log p_\theta(x) &= \log \int p_\theta(x|z)p_\theta(z) \frac{q_\phi(z|x)}{q_\phi(z|x)} dz \\
&\geq \underbrace{\int q_\phi(z|x) \log \left(\frac{p_\theta(x|z)p_\theta(z)}{q_\phi(z|x)} \right) dz}_{\text{EvidenceLowerBound(ELBO)}} \\
&= L(\theta, \phi; x) \\
L(\theta, \phi; x) &= \underbrace{\mathbb{E}_{q_\phi(z|x)} [\log p_\theta(x|z)]}_{\text{ReconstructionTerm}} - \underbrace{D_{KL}(q_\phi(z|x)||p_\theta(z))}_{\text{RegularizationTerm}}
\end{aligned}$$

1. The Reconstruction Term (The "Autoencoder" part):

- Math: $\mathbb{E}_{q_\phi(z|x)} [\log p_\theta(x|z)]$
- Meaning: This encourages the decoder to assign high probability to the original input x given the code z sampled from the encoder.
- In Practice: If $p(x|z)$ is Gaussian, maximizing log-likelihood is mathematically equivalent to minimizing Mean Squared Error. If Bernoulli, it is Binary Cross Entropy.

2. The Regularization Term (The "Variational" part):

- Math: $D_{KL}(q_\phi(z|x), p_\theta(z))$
- Meaning: This forces the encoder's output distribution $q_\phi(z|x)$ to look as close as possible to the prior $p(z)$ (the standard normal unit sphere).
- Why is this crucial? Without this term, the encoder would "cheat" by making $q_\phi(z|x)$ a delta function (a single point) effectively memorizing the data (like a standard autoencoder). The KL term forces the encoder to inject noise and smooth out the latent space, ensuring that valid new data can be generated by sampling from the prior.